

# Application of Fourier Neural Operator as an Autoregressive Neural Surrogate for Geothermal Simulation

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June 2, 2026

## Abstract

Geothermal energy is a promising solution to the worldwide growing demand for clean energy. Operating geothermal reservoirs requires careful engineering decisions, often involving repeated numerical simulation runs to identify optimal well controls. The expensive computational cost of these simulations has motivated interest in data-driven surrogate models, where a neural network is trained on simulation-generated data and acts as a proxy for future engineering workflows. Building such neural surrogates is a challenging problem: we want the model’s predictions to closely match the simulator outputs, and to maintain that accuracy autoregressively into the future, all while training on as little data as possible. In this work, we propose two 3D geothermal datasets to evaluate our surrogate model: one with homogeneous and one with heterogeneous porosity and permeability fields. Training on only 300 trajectories, our adapted Fourier Neural Operator with several architectural and training modifications achieves 0.1 % relative error on 3D pressure and temperature predictions, and remains stable autoregressively across 156 time steps.

## 1 Introduction

Numerical simulators have been used in geothermal reservoir management for a long time, supporting engineering workflows such as history matching, well placement, well-schedule optimization, uncertainty quantification, and production forecasting. Non-isothermal simulators include SHEMAT, FEFLOW, OpenGeoSys, ECLIPSE, and STARS, to name a few. While these tools differ in their discretization strategies, the governing equations must still be linearized, and the linear systems must be solved iteratively until convergence. This process becomes increasingly expensive as spatial resolution, physical complexity, and simulation duration grow. The cost is further amplified in engineering workflows such as optimal well control, where the simulation must be run hundreds of times to explore the decision space.

The prohibitive cost of repeated numerical runs has motivated growing interest in data-driven surrogate models, where a neural network is trained on trajectories generated by a simulator and serves as a fast proxy for iterative workflows.

[**TODO:** state the gap / what is missing in prior work, then the contributions.]

**Contributions.** In this work we:

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- release two 3D dual-porosity geothermal datasets (homogeneous and heterogeneous porosity/permeability) for benchmarking surrogate models;
- adapt the Local–Global Fourier Neural Operator (LoGlo-FNO) to 3D non-isothermal geothermal simulation with several architectural and training modifications;
- demonstrate 0.1 % relative error and stable 156-step autoregressive rollouts while training on only 300 trajectories;
- **[TODO: ablations / additional contributions.]**

## 2 Related Work

UNet-based models leverage convolutional encoders and decoders to learn multiscale spatial features, making them a natural fit for grid-based simulation data. Fourier Neural Operators (FNOs) and their variants approach the problem from a different angle: grounded in the theory of learning mappings between function spaces, they parameterize the solution operator directly in the frequency domain, offering resolution-invariant learning with strong theoretical foundations. Among these variants, the Local–Global Fourier Neural Operator (LoGlo-FNO) extends the standard FNO by explicitly learning high-frequency features through local convolutions alongside the global spectral modes, addressing a well-known limitation of purely Fourier-based approaches. Transolver generalizes the operator-learning paradigm further by attending over learned physical states rather than fixed spatial positions, offering flexibility for problems with complex geometries. Graph neural networks have also attracted attention, as they naturally represent the connectivity between grid blocks in a manner analogous to flux computation in discretized conservation laws.

Beyond architecture, training strategies also play a critical role in surrogate model quality. Autoregressive surrogate models are prone to error accumulation over long rollouts, and techniques such as the pushforward trick—which trains the model on its own predictions rather than solely on ground-truth inputs—have been shown to significantly improve temporal stability.

While the list above is far from exhaustive and the surrogate-modeling literature continues to grow rapidly, these approaches represent the principal architectural families relevant to our setting. In this work, we focus on architectures that operate on structured 3D grids, which rules out graph-based methods due to the prohibitive memory cost of constructing graphs over the large number of nodes in our discretization. We therefore benchmark against UNet, FNO, and Transolver as baselines, and build our primary model on the LoGlo-FNO architecture, introducing several modifications to adapt it to 3D non-isothermal geothermal simulation.

## 3 Methods

### 3.1 Dataset Generation

We generated two datasets: one homogeneous in porosity and permeability, and a heterogeneous counterpart. Each dataset contains 400 simulation runs, each spanning 156 weekly timesteps, corresponding to three years of operation. Because we use the dual-permeability (DUALPERM) setting in STARS, every run carries separate matrix and fracture properties. The reservoir spans 640 m in width, 320 m in length, and 160 m in height, discretized into  $10\text{ m} \times 10\text{ m} \times 10\text{ m}$  cells, giving  $64 \times 32 \times 16 = 32,768$  cells in total. The wells are arranged in a stacked inverted five-spot pattern, with 2 injectors and 7 producers, as shown in Figure 1. The porosity, permeability, and

[TODO: well pattern / domain figure]

Figure 1: Stacked inverted five-spot well pattern (2 injectors, 7 producers) and reservoir domain.

Table 1: Homogeneous-dataset rock properties.

|                   | Matrix | Fracture |
|-------------------|--------|----------|
| Porosity          | 0.05   | 0.005    |
| Permeability (mD) | 0.1    | 50       |

remaining simulation parameters are described in Section 3.1.1, and the 400 varying well schedules shared across both datasets are described in Section 3.1.2.

### 3.1.1 Porosity, Permeability, and Other Simulation Parameters

For the homogeneous dataset, porosity and permeability are held constant across the field (Table 1): the matrix has porosity 0.05 and permeability 0.1 mD, while the fractures have porosity 0.005 and permeability 50 mD.

For the heterogeneous dataset, we used unconditional Gaussian simulation to generate 400 matrix porosity realizations and 400 fracture porosity realizations (Table 2). Both use omnidirectional variograms. The horizontal variogram is spherical with 10 lags of 10 m, 150 m search radius, range 100 m, nugget 0.1, and sill 0.9. The vertical variogram is spherical with 5 lags of 10 m (one cell block each), 80 m search radius, range 50 m, nugget 0.1, and sill 0.9. Matrix porosity is drawn with mean 0.05 and variance  $2.5 \times 10^{-5}$ , and fracture porosity with mean 0.005 and variance  $1 \times 10^{-6}$ . To ensure every block sustains some flow, we clip matrix porosity to  $[0.035, 0.065]$  and fracture porosity to  $[0.002, 0.008]$ , corresponding to  $\pm 3$  standard deviations about each mean.

Permeability is then derived from porosity. Matrix permeability follows

$$k_{\text{matrix}} = \exp(29.182291 \phi - 4.017112) \text{ mD}, \quad (1)$$

which keeps matrix permeability within  $[0.05, 0.12]$  mD, and fracture permeability follows

$$k_{\text{fracture}} = \exp\left(3.21888 + 0.536479 \frac{\phi - \mu_\phi}{\sigma_\phi}\right) \text{ mD}, \quad (2)$$

which keeps fracture permeability within  $[5, 125]$  mD. Both datasets use a fracture spacing of 2 m, averaging about 5 fractures per cell block. These values were chosen so the simulation resembles a volcanic rock (porosity 0.05, permeability 0.1 mD) that has been mildly stimulated, mechanically or chemically—reflected in the dense fracturing and the elevated fracture permeability between 5 and 125 mD.

### 3.1.2 Well Control Schedules

We want our surrogate model to be robust and able to capture highly irregular and chaotic physical dynamics. With this goal in mind, we randomly divide the 156 timesteps into chunks of 2, 3, or 4 weeks. Within each chunk, we sample the rate for each injector from a uniform distribution between 500 and  $3000 \text{ m}^3 \text{ d}^{-1}$ . For the producers, we scale the sum of all injector rates by a scalar drawn from a uniform distribution between 0.92 and 1.05, ensuring the producers do not draw the reservoir down too quickly, and then randomly allocate this scaled total across the 7 producers. To make the

Table 2: Heterogeneous-dataset geostatistical parameters.

[**TODO:** fill variogram / distribution parameters table]

|               | Matrix               | Fracture           |
|---------------|----------------------|--------------------|
| Mean porosity | 0.05                 | 0.005              |
| Variance      | $2.5 \times 10^{-5}$ | $1 \times 10^{-6}$ |
| Clip range    | [0.035, 0.065]       | [0.002, 0.008]     |

dynamics even more irregular, every injector and producer has a 10 % chance of shutting in during each chunk, where its rate is set to zero and the remaining well rates are adjusted accordingly. This procedure yields highly irregular well-control schedules, which in turn drive complex, chaotic reservoir dynamics to test our surrogate model’s ability to learn and generalize with limited data.

### 3.2 Problem Formulation

After running the 400 simulations for each dataset, we extract the pressure and temperature fields at weekly intervals from the STARS output (.sr3) files, giving 156 time steps per trajectory. Each field has four channels: matrix temperature, fracture temperature, matrix pressure, and fracture pressure. We split the 400 trajectories into 300 for training and 100 for testing.

The surrogate model’s task is to predict the field at the next time step from the current field and the current well-control action. From each simulation we have a trajectory of states sampled at regular weekly intervals, and this temporal discretization motivates us to view the problem as an Euler discretization of the reservoir dynamics, where an operator  $\mathcal{G}_\theta$  models the per-step rate of change,

$$\frac{\mathbf{u}_{t+1} - \mathbf{u}_t}{\Delta t} = \mathcal{G}_\theta(\mathbf{u}_t, \mathbf{a}_t). \quad (3)$$

Taking a unit step size  $\Delta t = 1$ ,

$$\mathbf{u}_{t+1} = \mathbf{u}_t + \mathcal{G}_\theta(\mathbf{u}_t, \mathbf{a}_t), \quad (4)$$

where  $\mathbf{u}_t = [T_{\text{matrix}}, T_{\text{fracture}}, P_{\text{matrix}}, P_{\text{fracture}}]$  is the four-channel state field at time  $t$ ,  $\mathcal{G}_\theta$  is the surrogate model, and  $\mathbf{a}_t$  is the well-control action. The action is encoded as a  $16 \times 64 \times 32$  tensor that holds each well’s rate at its perforated cells (layers 2–13, see Table 1) and zero elsewhere. We concatenate it with a second tensor of the same shape: a binary mask that is 1 at active perforated cells and 0 at shut-in cells. Note that this is equivalent to having a clean residual connection from  $\mathbf{u}_t$  to  $\mathbf{u}_{t+1}$ , which we implement our LOGLO-FNO adaptation and all of the baselines.

Beyond predicting the next state  $\mathbf{u}_{t+1}$ , the same setup can be extended to learn any derived quantity of interest. Formally, given the current and next states we predict a target vector

$$\mathbf{q}_t = \mathcal{H}_\psi(\mathbf{u}_t, \mathbf{u}_{t+1}) = \mathcal{H}_\psi(\mathbf{u}_t, \mathbf{u}_t + \mathcal{G}_\theta(\mathbf{u}_t, \mathbf{a}_t)), \quad (5)$$

where  $\mathcal{H}_\psi$  is a separate auxiliary head. The choice of  $\mathbf{q}_t$  is problem-specific and can be set to whatever a downstream workflow needs. Here we let  $\mathbf{q}_t$  be the bottomhole pressure at each of the 9 wells and the energy production rate at each of the 7 producers (16 values in total).

### 3.3 Model Architecture

[**TODO:** Describe the adapted LoGlo-FNO: three parallel branches per block (global spectral, local patch-based spectral, high-frequency MLP), AdaLN-Zero action conditioning, zero-init projection for identity-at-init, patch size (8, 8, 8). Describe the auxiliary head predicting per-well BHP and energy production rates. Include an architecture diagram.]

[**TODO:** architecture diagram]

Figure 2: Model architecture.

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**Algorithm 1** Pushforward autoregressive training (sketch)

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1: [**TODO:** pseudocode for the training loop]

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### 3.4 Training

[**TODO:** Pushforward training (unroll  $k$  AR steps,  $k$  schedule), multi-loss objective (weighted MSE, H1/Sobolev, radial-binned spectral, mass-balance-equation residual, mean-field pressure, auxiliary MSE), adaptive input noise, EMA, LR schedule (linear warmup  $\rightarrow$  cosine decay), fp32 training.]

## 4 Experiments

### 4.1 Experimental Setup

[**TODO:** Hardware (H100), optimizer, batch size, epochs, seeds, normalization (min-max; log1p for energy rate), reproducibility details.]

### 4.2 Baselines

[**TODO:** UNet3D, FNO, Transolver configurations and the experiment matrix.]

### 4.3 Evaluation Metrics

[**TODO:** Relative  $L_2$  / per-channel relative error, rollout error vs. horizon, spectral error, auxiliary (BHP, energy rate) error.]

## 5 Results

### 5.1 Homogeneous Reservoir

[**TODO:** quantitative table + rollout-error-vs-time plot + qualitative fields.]

### 5.2 Heterogeneous Reservoir

[**TODO:** same structure as homogeneous; emphasize generalization across static fields.]

### 5.3 Ablations

[**TODO:** pushforward on/off, individual loss terms, EMA, noise injection, local-global branches.]

## 6 Discussion

[**TODO:** Interpretation, failure modes, limitations (data regime, single reservoir geometry), and implications for well-control optimization.]

Table 3: Test-set accuracy on the homogeneous dataset (lower is better).

| Model            | Rel. $L_2$ (%) | 156-step Rel. $L_2$ (%) | Aux. error | Params |
|------------------|----------------|-------------------------|------------|--------|
| UNet3D           |                |                         |            |        |
| FNO              |                |                         |            |        |
| Transolver       |                |                         |            |        |
| LoGlo-FNO (ours) |                |                         |            |        |

## 7 Conclusion

[**TODO:** Summarize contributions and outline future work.]

## Acknowledgments

[**TODO:** funding, compute (DRAC/Fir cluster), collaborators.]

## Data and Code Availability

[**TODO:** links to datasets and repository.]

## References